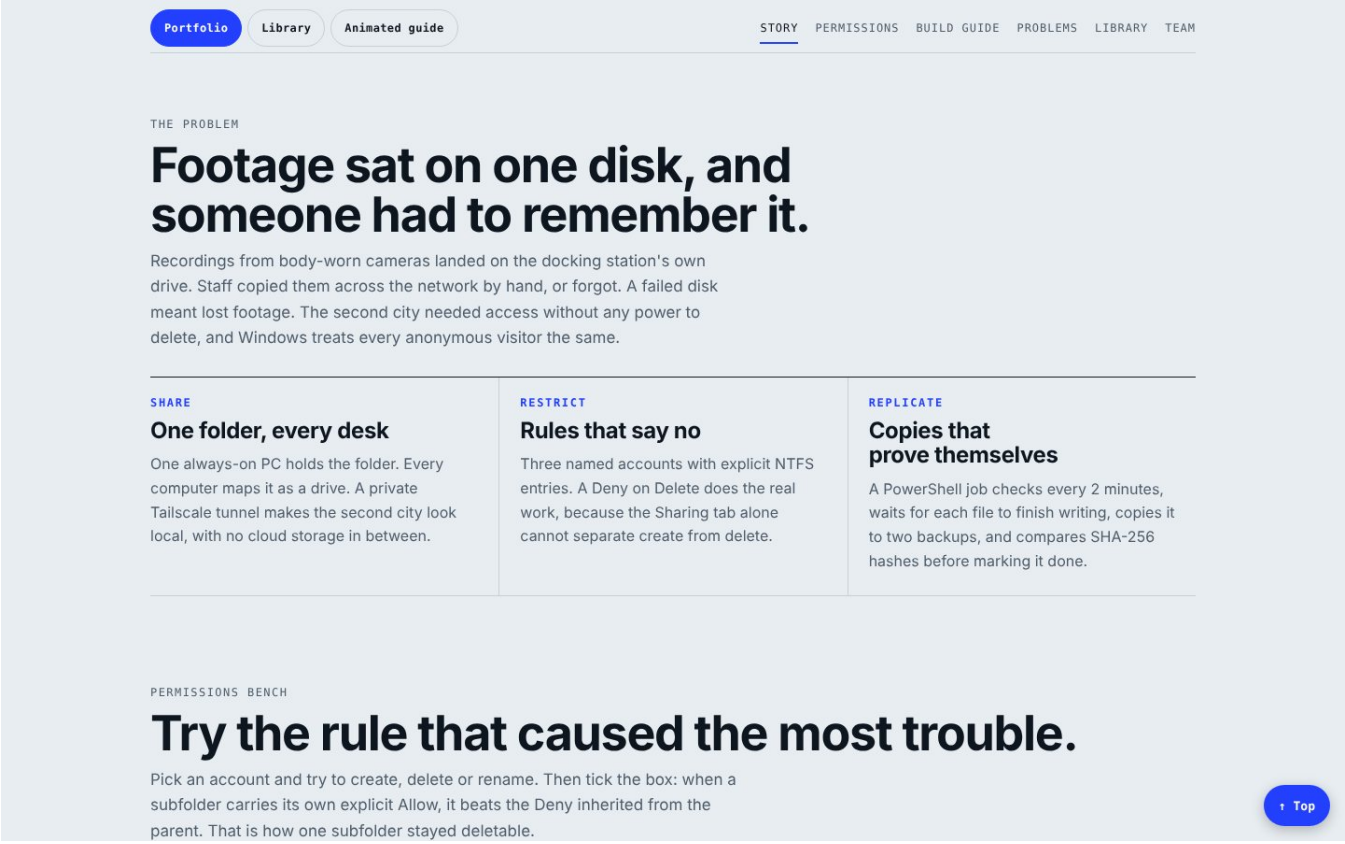
PROJECT README

BC680 Body-Cam Data Backup System

Three hash-verified copies of every body-camera recording across two offices, checked every two minutes.

Type	IT systems and automation
Organisation	Delivered through Expert Mobile Communications Ltd.
Year	2026
Role	Designer, builder and documentation author
Repository	github.com/Masum-jubayel/Bc680-Bodycam-Data-Backup
Live site	masum-jubayel.github.io/Bc680-Bodycam-Data-Backup



Project site overview.

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1. What the project is

A shared-drive and backup system for BC680 body-camera recordings. Each recording is fingerprinted with SHA-256, locked in as the master and copied to two backups in a main office and a remote office. Copies are verified every two minutes. The project includes a rebuild guide, animated staff training, a customer presentation, PDF guides and an explainer video.

2. Why it was built

Body-camera recordings are evidence. They must never be lost, silently changed or deleted, and staff in two offices need simple, reliable access.

3. How it works, step by step

Step	Stage	What happens
1	**Capture**	Recordings land on the primary PC in the main office.
2	**Fingerprint**	A SHA-256 hash is recorded and the file becomes the master.
3	**Copy**	The file is copied to the remote-office backup over a Tailscale tunnel, and optionally to OneDrive.
4	**Verify**	The hash of every copy is checked again; any difference is logged.
5	**Protect**	Backup folders use NTFS Deny Delete; even the backup account cannot remove files.
6	**Access**	Staff open recordings through a mapped drive (Z:) with read-only accounts.

4. Features

- Three verified copies of every recording
- Verification every 2 minutes via Task Scheduler
- Least-privilege accounts: OfficeStaff, RemoteStaff, BackupWriter
- Encrypted site-to-site link with Tailscale
- Twelve documented problems with fixes and a 55-step rebuild guide
- Six-chapter animated training guide
- 13-slide customer presentation, five PDF guides and a 58-second video

5. Architecture

Main office primary PC (master) and remote office backup PC connected by Tailscale; SMB shares with NTFS permissions; PowerShell scripts run by Task Scheduler; optional OneDrive off-site copy.

6. Technologies used

Each technology and the job it does in this project:

Technology	What it is used for
Windows	Operating system on the office PCs that store and serve the recordings.
PowerShell	Windows scripting language used to automate copying, hashing, verification and scheduled tasks.
SHA-256	Cryptographic hash: a unique fingerprint of a file. If even one bit changes, the fingerprint changes, proving whether copies are identical.
SMB / NTFS permissions	SMB shares folders over the network; NTFS permissions control exactly who may read, write or delete.
Tailscale (WireGuard)	Encrypted private network between the two offices, without opening ports to the internet.
OneDrive	Microsoft cloud storage used as an optional off-site copy.
Task Scheduler	Windows service that runs the verification script every two minutes.
HTML5 / CSS3	HTML structures each page; CSS handles layout, colour, typography and responsive design for phones and desktops.
JavaScript	The language of the browser; powers the interactive parts of the interface.
SVG animation	Scalable vector graphics animated with CSS/JavaScript for the training guide diagrams.
GitHub Pages	Free static hosting directly from a GitHub repository.

7. Security & privacy

- No internet-exposed ports; traffic runs over WireGuard.
- Deny Delete on backups.
- Hashes prove integrity of every copy.

8. Results

- Triple-copy, hash-verified storage.
- Deletes refused on backups.
- Complete hand-over package for the customer.

9. Next steps

- Alerting by email when a verification fails

Author: Masum Jubayel - Network, Security & IT Engineer. Portfolio: masum-jubayel.vercel.app | GitHub: github.com/Masum-jubayel